

pH and the pH Scale

pH, pOH and calculations for strong acids and bases

Name _____ Class _____ Date _____

USEFUL RELATIONSHIPS (25 °C)

$$\text{pH} = -\log[\text{H}_3\text{O}^+] \quad \text{pOH} = -\log[\text{OH}^-] \quad \text{pH} + \text{pOH} = 14.00 \quad [\text{H}_3\text{O}^+] = 10^{-\text{pH}}$$

Give pH values to two decimal places.

A. Understanding

1. At 25 °C, classify each pH as acidic, neutral or basic: 2.5, 7.0, 9.8, 6.9.

2. How many times greater is $[\text{H}_3\text{O}^+]$ in a solution of pH 2 than in a solution of pH 5?

B. Strong acids

3. Calculate the pH of 0.0010 M HCl.

4. Calculate the pH of 0.25 M HNO_3 .

C. Strong bases

5. Calculate the pH of 0.010 M NaOH.

6. Calculate the pH of 0.0050 M barium hydroxide, Ba(OH)₂.

D. Concentration from pH

7. Calculate [H₃O⁺] for a solution with (a) pH 3.20 and (b) pH 11.50.

8. What is [OH⁻] in a solution with pH 5.00?

E. Challenge

9. 0.10 M HCl is diluted 100 times with water. What is the new pH? By how much has the pH changed?